

FINAL PROJECT REPORT: SIMULATION LABORATORY

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1 Introduction

The project is to model, design and implement a robust control algorithm for hydraulically driven crane. The crane consists of a complex mechanical components such as a weigh attached movable lift boom attached and double-acting hydraulic actuator. Th primary technical challenge is to integrate a multi-domain subsystems: mechanical rigid body dynamics, structural flexibility and nonlinear fluid power into a high fidelity simulation environment. The ultimate goal is to achieve precise motion control programmed to lift the boom to mimic realistic movement of the actual experiment therefore it requires a deep understanding of real-world laboratory system before numerical implementation.

2 Method

The methodology for this project followed a structured four part engineering guidelines provided by Instructor to transition from a physical laboratory crane to a high fidelity digital twin. The structured four part engineering are

- Mechanical Structure and Multibody Modeling
- Hydraulic Actuation System
- Structural Flexibility Analysis
- Control Algorithm Design

These are achieved using computer aided software as such SolidWorks for modeling and modal analysis, and MATLAB for implementing the custom component coding and structural flexibility PDE analysis, MATLAB Simulink and Simscape Multibody for simulation of the model.

2.1 Mechanical Structure and Multibody Modeling

2.1.1 Mechanical Structure

The overall objective is to construct parts in SolidWorks and create a working model in MATLAB Simulink and Simscape Multibody. The project requires to meet a specific specification, technical requirements and tool. The parts are modeled with the help of SolidWorks 2025 to fabricate the designs given specifications. The hydraulic system and point mass are designed in MATLAB Simulink and Simscape Multibody 2024. As shown in the Figure 1 given design from the project scope, the task was to fabricate the same working model.

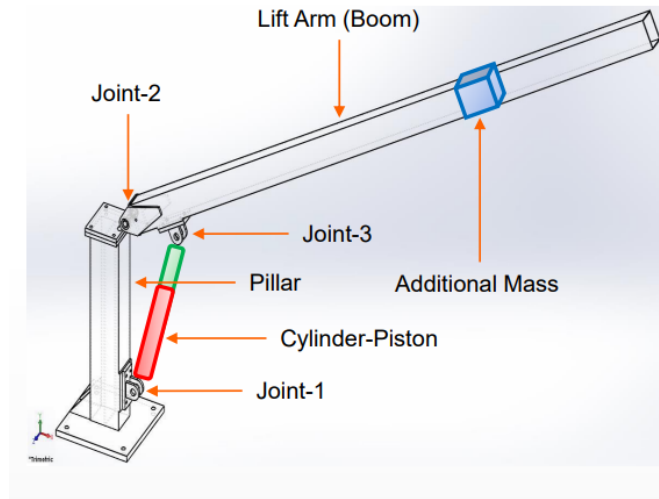


Figure 1: Working Model given by the project

- Material properties were assigned to all components using steel density of 7850 kg/m^3 [34]. Mass properties, center of mass and moment of inertias were calculated in SolidWorks and validated against values obtained in MATLAB Simulink. Before exporting to STL, the unit consistency was ensured to set MKS system.
- It was also necessary to satisfy specific joint coordinate requirements [3]:
 - Joint-1: $[X, Y, Z] = [0, 0, 0] \text{ mm}$
 - Joint-2: $[X, Y, Z] = [-33, 880, 0] \text{ mm}$
 - Joint-3: $[X, Y, Z] = [278, 727, 0] \text{ mm}$
 - End of Lift Arm: $[X, Y, Z] = [4167, 880, 0] \text{ mm}$
- Simulation Environment: Simscape Multibody (MATLAB/Simulink) serves as the primary engine for mechanism construction and dynamic simulation. The properties of the part must be same as the imported CAD model files.
- System Constraints: The final mechanism must have one degree of freedom and contain no redundant constraint equations.

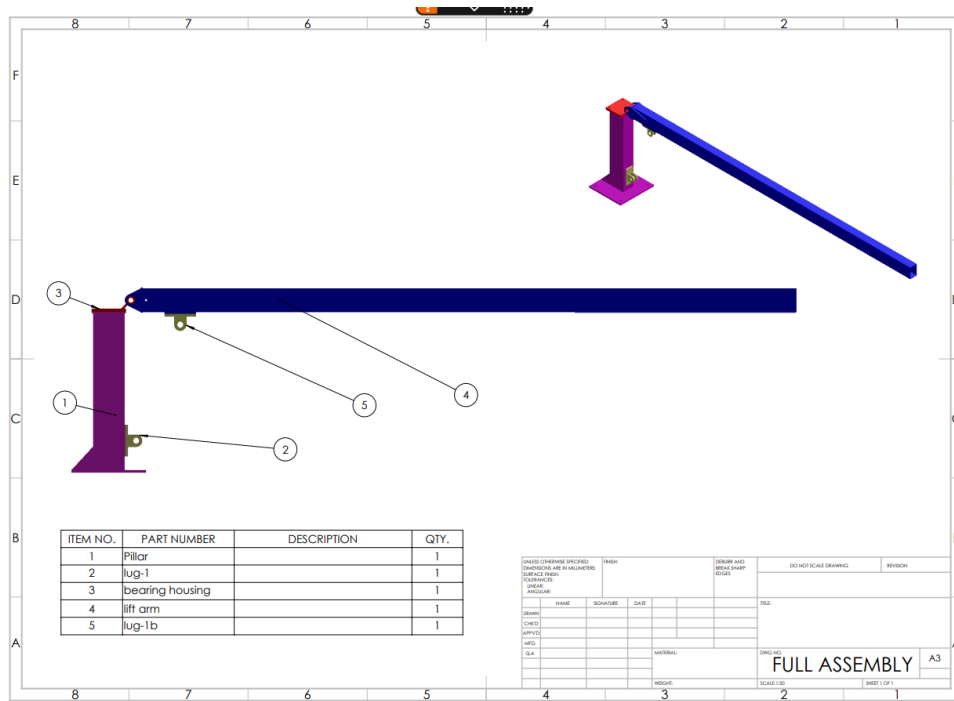


Figure 2: Bill of Material

The components to be designed in SolidWorks consist of five parts, one of which—Lug-1—is duplicated. Figure 2 provides the bill of materials. For simulation, the system was configured with appropriate joint connections and reference frames. A sinusoidal input was applied to create the motion and the gravity was defined along the negative Y-axis. The simulation model was then analyzed the dynamic behavior of the crane system.



Figure 3: New Coordinates at the Joints

2.1.2 Simulation Model

The mechanical components are imported individually into the MATLAB Simulink and Simscape Multi-body using STP file extension. The schematic of the simulation model is set up accordingly and new ports are set up in each files in order to streamline the assembly in the simulation.

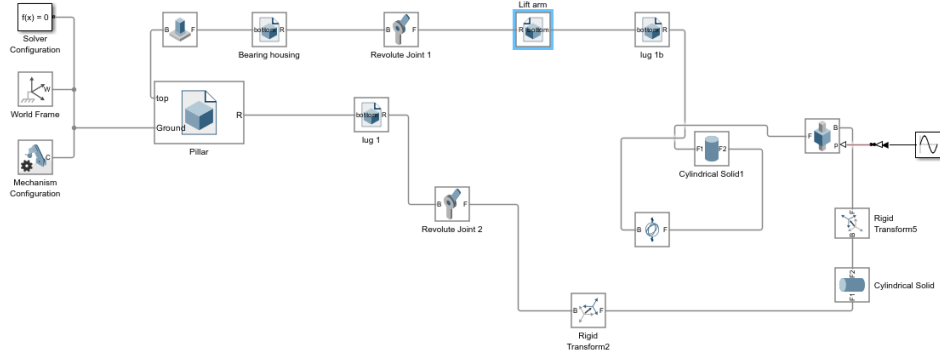


Figure 4: Schematic of the Simulation

The assembled model is simulated and using the Sinewave function. The components are assembled in a way that Y axis is Vertical, X is Horizontal and Z is out of the plane. The gravity is also switched to Y axis negative position. Watch Video Analysis : The video of the motion is provided through this link.

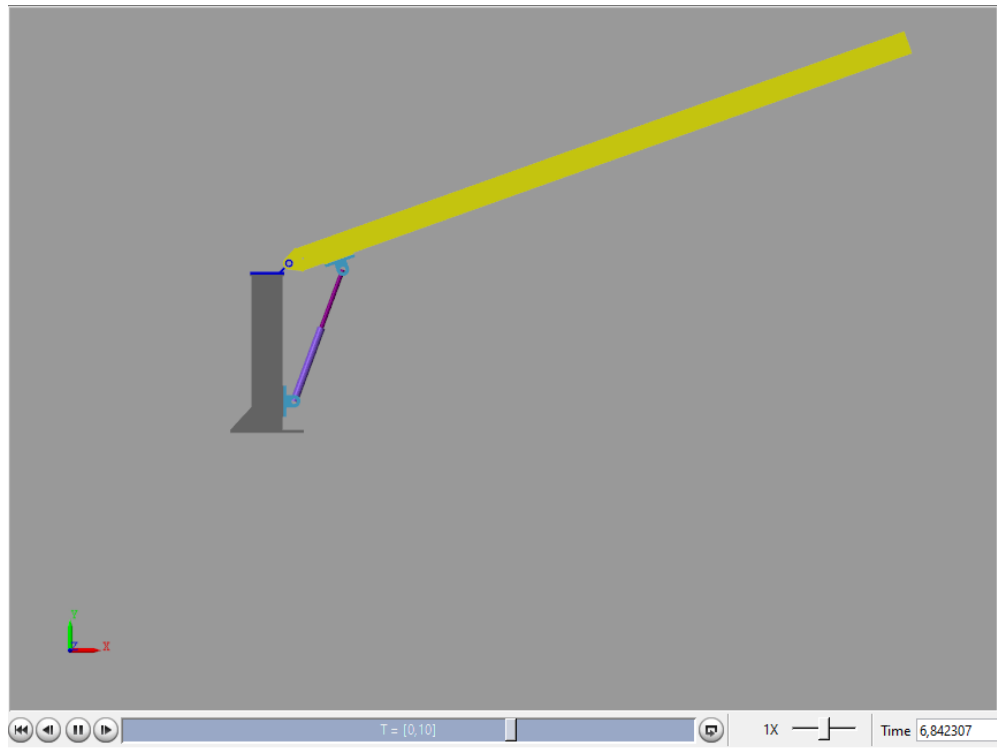


Figure 5: Simulation of the model

2.2 Hydraulic Actuation System

The hydraulic system was model in MATLAB Simulink and Simscape using the custom component and lumped fluid theory. The implementation focuses on actuator behavior and system-level hydraulic integration.

2.2.1 System Specification

1. Hydraulic Actuator (Double-acting Cylinder)

- Piston Bore Diameter: 50 mm.
- Piston Rod Diameter: 30 mm.
- Total Stroke Length: 300 mm.
- Mechanical Efficiency (η): 0.9 (unitless).
- Bulk Modulus of Cylinder (B_c): 21,000 MPa.

2. Hydraulic Pump & Valve

- Pump Model: Parker PV016.
- Initial Testing Pressure: 150×10^5 Pa (assume 150 bar).
- Input Shaft Rotation Speed: 1500 rpm.
- Control Valve Code: BOSCH 0811404611.
- Hose Volume (Pump to Directional Valve): 3×10^{-3} m³.

3. Fluid & Conduit Properties

- Fluid Density (ρ): 850 kg/m³.
- Kinematic Viscosity (ν): 32×10^{-5} m²/s.
- Bulk Modulus of Oil (B_0): 1500 MPa.
- Connecting Hose Length: 1.5 m.
- Hose Inner Diameter: 3/8".
- Relative Amount of Trapped Air: 0.4.

4. Mathematical Modeling Constants

- Hose Bulk Modulus (B_h): Calculated as

$$B_h = 10^6 \cdot \sqrt{p_h}$$

where p_h is the pressure in Pascals.

- Lumped Fluid Theory Reference: Effective bulk modulus (B_e) and volume flow rate equations utilize the volumes and areas derived from the diameters above.

The scope of this report covers the mathematical modeling and simulation setup of the hydraulic drive system. The boundaries of this phase include:

- **Custom Component Development:** Designing S-function or Simscape (.ssc) models for the directional valve and double-acting cylinder using lumped fluid theory.
- **Hydraulic Circuit Integration:** Connecting the modeled pump, tank, hoses, and valves to represent the full energy transmission path.
- **Static Equilibrium Analysis:** Utilizing the "Steady State Manager" to determine the initial pressure conditions required for stable simulation start-up.
- **Laborious Modeling:** Focusing on the most time-intensive part of the crane's design—the hydraulic actuators—prior to control logic implementation.

2.2.2 Directional Control Valve

The directional control valve was modeled as a custom Simscape component using an .ssc file. The valve includes four hydraulic ports (P, T, A, and B) and one input signal U_{in} . These represents as the spool control command. In order to make valve response more realistic, a first-order dynamic is added using the time constant τ , so the internal valve opening U follows the input signal gradually instead of changing instantly. The pressure differences between the ports are calculated inside the component, and these pressure drops are used to compute the mass flow rates through the valve. The flow equations are based on a semi-empirical model, where the flow rate is proportional to the control signal and to $\sqrt{|p|}$, while the sign of the pressure difference defines the flow direction. When $U > 0$, the valve allows flow from P to A and from B to T. When $U < 0$, the flow path changes to P to B and from A to T. In the neutral position ($U = 0$), all flow paths are closed, meaning no flow passes through the valve. The specification is configured as provided. The SSC file for the code can be viewed in Appendix B.1

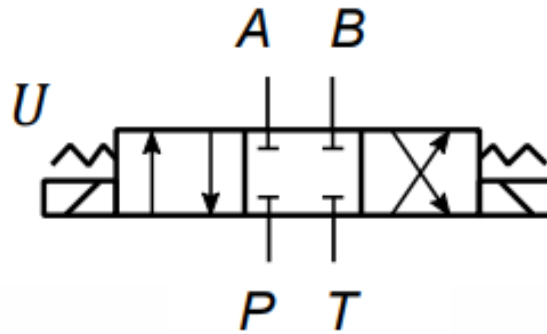


Figure 6: Directional Control Valve

2.2.3 Double-acting Hydraulic Cylinder

This custom Simscape component models a double-acting hydraulic cylinder using lumped fluid theory. The block contains two isothermal liquid ports (A and B), which represent the two chambers of the cylinder. The piston position x and piston velocity $\dot{x}$ are provided as input signals, and the output of the component is the generated actuator force F . The model calculates the chamber volumes based on the piston position and includes a minimum dead volume V_0 to avoid numerical issues when the piston approaches the stroke limits. The pressure dynamics are calculated using effective bulk modulus values, which combine the compressibility of the hydraulic oil, the flexibility of the cylinder walls, and the effect of trapped air in the hoses. Mass flow rates are then computed from the pressure derivatives and the piston motion term, meaning the flow depends both on fluid compression and on volume changes caused by piston movement. Finally, the actuator force is computed from the pressure difference across the piston areas. The following code for the part is provided in Appendix B.2

2.2.4 Lumped mass fluid theory

The hydraulic actuator is modeled as a two-chamber lumped parameter system, consisting of a rod-side chamber (A) and a cap-side chamber (B). Each chamber is treated as a single compressible control volume

with spatially uniform pressure. The total volume of each chamber includes both the variable cylinder volume and an additional constant dead volume V_0 representing the lateral pipe connections. The chamber volumes are therefore defined as

$$V_A = \max(A_1 x, V_0), \quad V_B = \max(A_2(l - x), V_0), \quad (1)$$

where A_1 and A_2 are the effective piston areas, x is the piston displacement, and l is the total cylinder length. The mass balance for each compressible chamber follows from the lumped continuity equation,

$$\dot{m} = \rho \left(\frac{V}{B_{\text{eff}}} \dot{p} + \dot{V} \right), \quad (2)$$

which accounts for both fluid compressibility and moving boundary effects. Thus, the mass flow rates at ports A and B are given by

$$\dot{m}_A = \rho \left(\frac{V_A}{B_{e2}} \dot{p}_A + A_1 \dot{x} \right), \quad (3)$$

$$\dot{m}_B = \rho \left(\frac{V_B}{B_{e3}} \dot{p}_B - A_2 \dot{x} \right), \quad (4)$$

where B_{e2} and B_{e3} are the effective bulk moduli that incorporate oil compressibility and structural compliance. The resulting actuator force is obtained from the pressure difference across the piston,

$$F = A_1 p_A - A_2 p_B, \quad (5)$$

with additional mechanical losses modeled through an efficiency factor.

2.2.5 Schematic of MATLAB

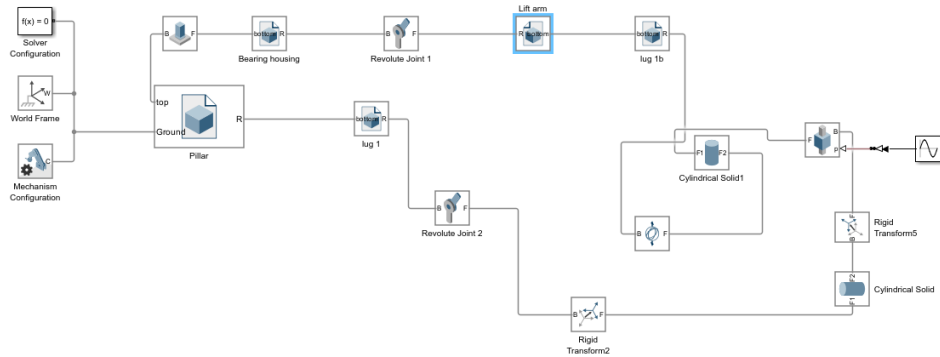


Figure 7: Hydraulic, Directional Valve and Pump Schematic

The link to the animation of the working mechanism is here: [Video of Analysis](#). Apparently there is not a lot of actuation which is to be fixed in upcoming report. The pump parameters values are obtained from the Parker PV016.

- Nominal Pressure: 150 *bar*
- Nominal Displacement: 0.3 m^3/rad
- Nominal Shaft Angular Velocity 1500 *rpm*

These numbers are the constant values obtained from the catalog. It establish the baseline flow rate of the system as the pump delivers flow to port P on this driving shaft rotation.

2.3 Structural Flexibility Analysis

2.3.1 Pressure Compensated Pump Integration

The pressure compensated pump was not incorporated in the Simulink model. In this work, this module is first integrated before proceeding to the structural flexibility objectives. The schematic of the integration is shown below, and the required pump data are inserted based on the reference catalog.

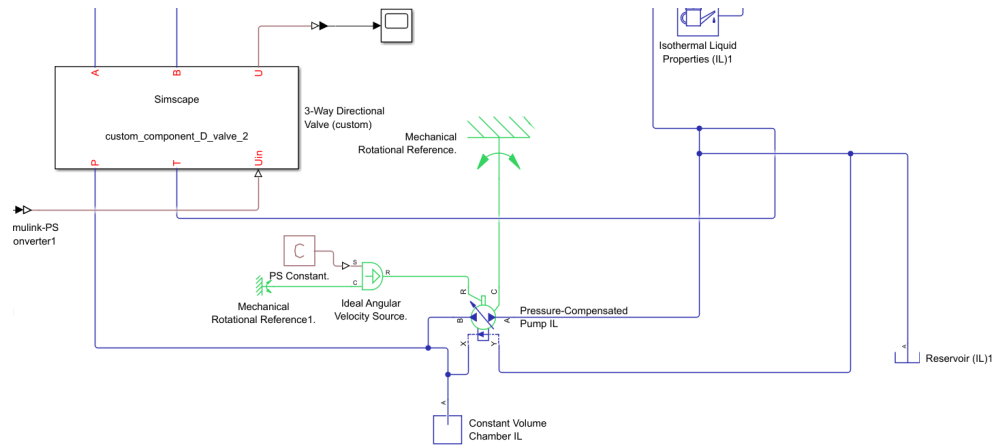


Figure 8: Pressure Compensated Pump Integration

Using the pump reference data, the following data are inserted.

NAME	VALUE
▼ Pump	
Leakage and friction parameterization	Analytical ▼
> Nominal displacement	16 cm^3/rev ▼
> Nominal shaft angular velocity	1500 rpm ▼
> Nominal pressure gain	350 bar ▼
> Volumetric efficiency at nominal conditi...	0.92
> No-load torque	0 $N*m$ ▼
Check if pressures are less than pump ...	None ▼
> Mechanical efficiency at nominal condit...	0.88
> Pressure Compensation	

Figure 9: Data required for Pressure Compensated Pump

2.3.2 SolidWorks Remodeling and Modal Analysis

The lift boom CAD model is remodeled to include Lug-1 and pins within the same CAD file by creating another configuration. The reason to combine all components into a single CAD file instead of using an assembly is to overcome tolerance issues that are required to be solved in MATLAB. The model also includes a plate where the mass will be attached, with the location depending on the student number, which is 3600 mm in this case.

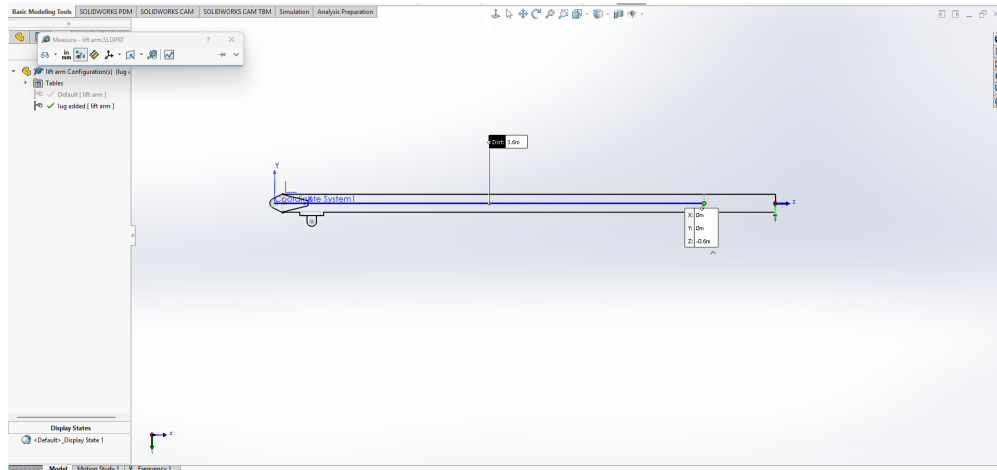


Figure 10: Plate attachment in New CAD model

The model is assigned stainless steel material, and modal analysis is conducted using 50 mode shapes. The constraints are applied at the pin locations where revolute joints occur in order to mimic the MATLAB Simulink dynamics.

2.3.3 MATLAB: Reduced Order Element

To implement the reduced order element framework in MATLAB, the CAD model is exported as an STL file with the option to avoid translation to positive space. The model is then converted into proper MKS units. The coordinates of the origins corresponding to joint locations (Joint 1 and Joint 2) and the additional mass attachment point are defined. It is important that the sequence of the origins matrix and face IDs are correctly assigned.

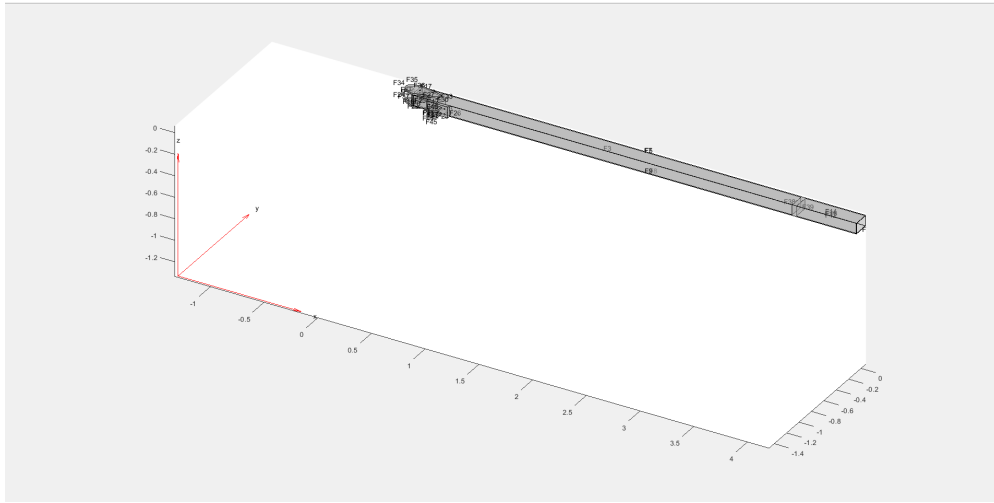


Figure 11: FaceIDs of the model

The meshed body of the liftboom. Red, Green and Blue represent the places where attachment will be made.



Figure 12: Meshed Model

Material properties, including Young's modulus, Poisson's ratio, and density, are defined, and the geometry is imported into the PDE model. A mesh is generated, and boundary conditions are applied at selected face IDs using multipoint constraints. The reduced order model is then generated within the frequency range of 0 to $1e3$. The stiffness, mass, and damping matrices are extracted, and modal damping is included Appendix: Reduced Order MATLAB CODE: B.3.

2.3.4 Experimental Data Processing

The cleaned data is then imported into The experimental data corresponding to the student number (70 kg, 3600 mm, 150 bar) is imported from a CSV file and converted into XLSX format using Excel. The data is cleaned by removing delimiters and deleting unnecessary information above the numerical dataset. The numerical values are reformatted to ensure compatibility with MATLAB, as the original format is interpreted as text. Decimal points are converted appropriately, and only relevant columns are retained.

The first column is used as time, and the third column represents the control input U_{in} . Simulink using the *From Spreadsheet* block, where the first column is treated as time and the output data type is set to double.

2.3.5 Simulink: Reduced Order Flexible Solid

The flexibility of the lift arm is implemented in Simscape Multibody using the Reduced Order Flexible Solid block. Instead of defining stiffness, mass, and damping manually, the reduced order model generated in MATLAB is directly imported. The STL geometry and material properties are included, and the interface frames appear as ports corresponding to the reference origins defined earlier. The stiffness, mass, and damping matrices obtained from the reduced model are inserted into the block to represent the flexible behavior of the structure.

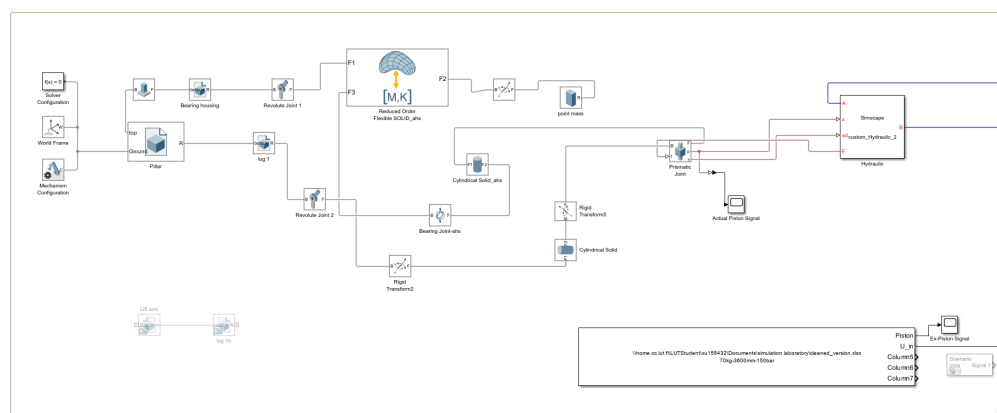


Figure 13: Schematic of Reduced Order Solid Block

Reduced Order Flexible Solid

☒ Auto Apply

Settings	Description	VALUE
NAME		
VALUE		
▼ Unit System		
Unit System	SI (m, kg, s)	
▼ Interface Frames		
Number of Frames	3	
Origins	arm.P	<3x3 double
Orientations	repmat([1 0 0 0], 3, 1)	<3x4 double
▼ Reduced Order Matrices		
Stiffness Matrix	arm.K	<20x20 double
Mass Matrix	arm.M	<20x20 double
▼ Damping		
Type	Damping Matrix	
Damping Matrix	arm.C	<20x20 double
▼ Fidelity		
Reduction	None	
▼ Graphic		
Type	Partitioned Geometry	
File Name	\\home.cc.lut.fi\LUTStudent\xu155432\Documents\simulation la...	
Unit Type	From Unit System	
> Visual Properties	Simple	

Figure 14: Detailed imported information

2.4 Control Algorithm Design

The control system was designed as a closed-loop feedback mechanism to simulate the crane's vertical displacement. The core of the system is the PID Controller which continuously calculates the error between the desired output (target position of the provided data) and the actual output from the simulation.

Mathematical Implementation

The controller applies "correction" signal to the directional control valve based on three distinct terms:

- Proportional (K_p): is used to decrease the rise time for boom to reach 100mm target quickly.
- Integral (K_i): accumulates the error log to eliminate the steady-state error ensuring the arm to reach the required height.
- Derivative(K_D): anticipates the future behavior to reduce overshooting and settling time which is critical for stabilizing the flexible lift arm.

The total output of the controller is represented as:

$$K_p + \frac{K_I}{s} + K_D s$$

Simulation Implementation

The simulation implementation initially uses a constant reference input to tune and verify the PID controller before attempting to track the complex signal from the experimental .x/sx file.

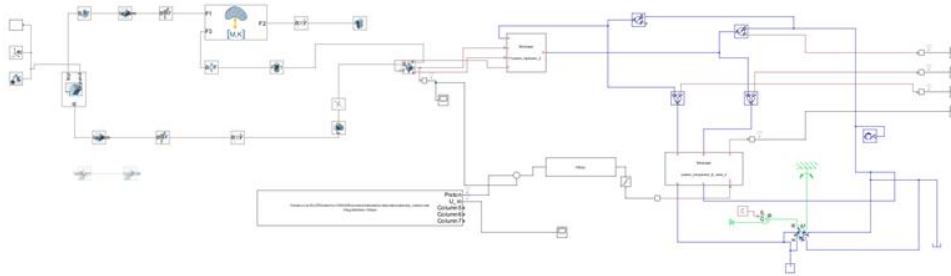


Figure 15: Schematic with PID

- Phase 1: Tuning with a constant "step" input. Before importing laboratory data, the step block in Simulink to represent the desired output.
 - This allows to verify the system is working and to tune K_p, K_I, K_D to achieve a stable response.
- Phase 2: once the controller is tuned and system responds correctly to the constant step, the step block is replaced with "From Spreadsheet" Block.
 - It imports the "Piston Position" column from the Excel File to serve as a new dynamic reference.

The tuning satisfied the PID requirements by bringing the actual output significantly closer to the desired reference position compared to the untuned open-loop response. It is to ensure K_p, K_I, K_D values not to cause the control signal to stay saturated at (+ or -)10V for too long.

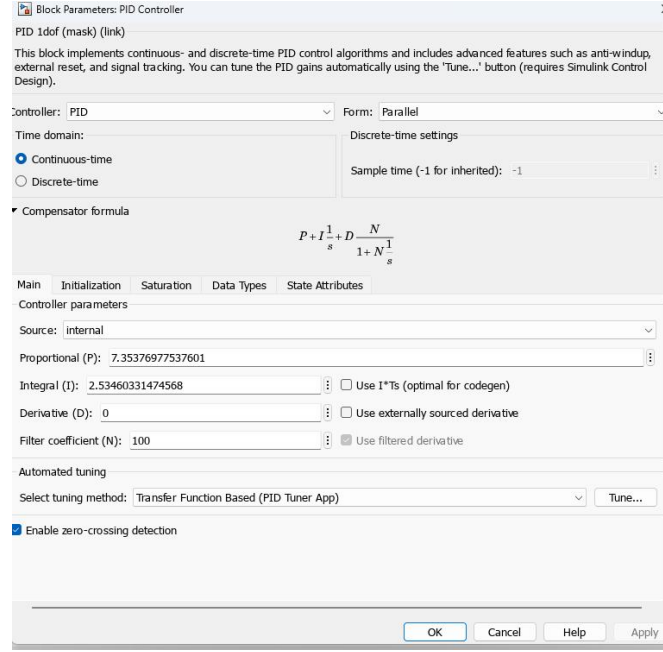


Figure 16: PID tuned configuration

3 Results

For the mid-report part 1, the goal is to achieve the working model in the simulation of MATLAB. Then to obtain the DOF of the working model to be 1 and to make sure there is no redundancy. The DOF is obtained using the Statistics Viewer of MATLAB debugging tool.

Description	Value
Number of bodies (total, excluding ground)	4
Number of rigidly connected components (excluding ground)	4
Number of flexible bodies	0
Number of joints (total)	5
Number of explicit tree joints	4
Number of implicit 6-DOF tree joints	0
Number of cut joints	1
Number of constraints	0
Number of tree degrees of freedom (total)	3
Number of tree joint degrees of freedom	3
Number of flexible body degrees of freedom	0
Number of position constraint equations (total)	2
Number of position constraint equations (non-redundant)	2
Number of mechanism degrees of freedom (minimum)	1
State vector size	12
Average number of degrees of freedom in kinematic loops	3

Figure 17: Degree of Freedom in Statistic View

3.1 Modal Analysis

Table 1: Selected Natural Frequencies and Periods

Mode No.	Frequency (rad/s)	Frequency (Hz)	Period (s)
1	0	0	∞
2	0	0	∞
3	0	0	∞
4	0.00041	6.57×10^{-5}	15221
5	0.00864	0.00137	727.32
6	0.02160	0.00344	290.91
7	215.23	34.25	0.0292
8	293.22	46.67	0.0214
9	632.78	100.71	0.00993
10	857.98	136.55	0.00732
11	1236.2	196.75	0.00508
12	1592.7	253.49	0.00394

The modal analysis is performed to identify the natural frequencies and mode shapes, which describe how the system deforms or vibrates corresponding to its natural frequency. This provides an understanding of structural stability, resonance behavior, and dynamic response patterns.

The first three mode shapes represent rigid body motion, as there is no excitation. Out of the 50 computed modes, only a limited number are relevant for the simulation. High-frequency modes that are not relevant are neglected. The selection of mode shapes is based on identifying the relevant direction of deformation, particularly in the XY plane, since the constraints are not fully defined.

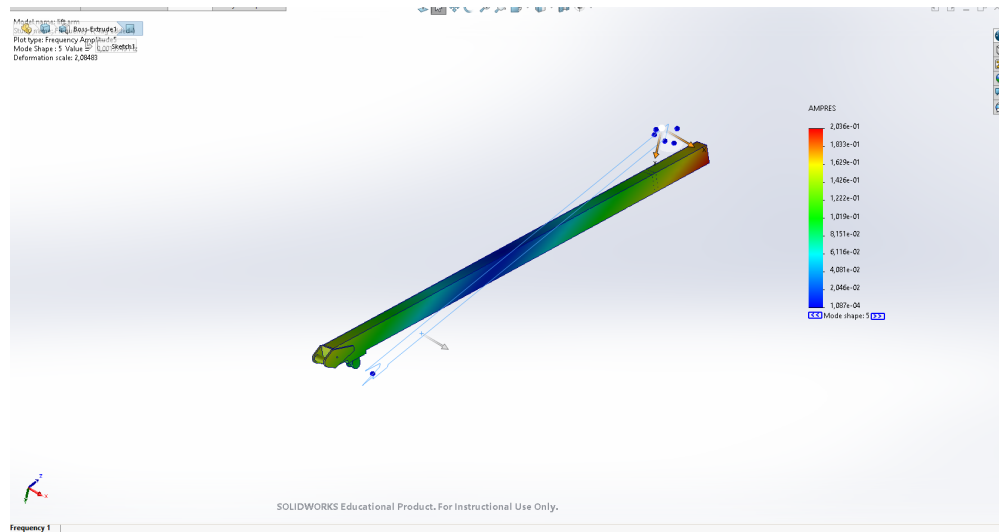


Figure 18: Mode 5 with 0,00137 Hz

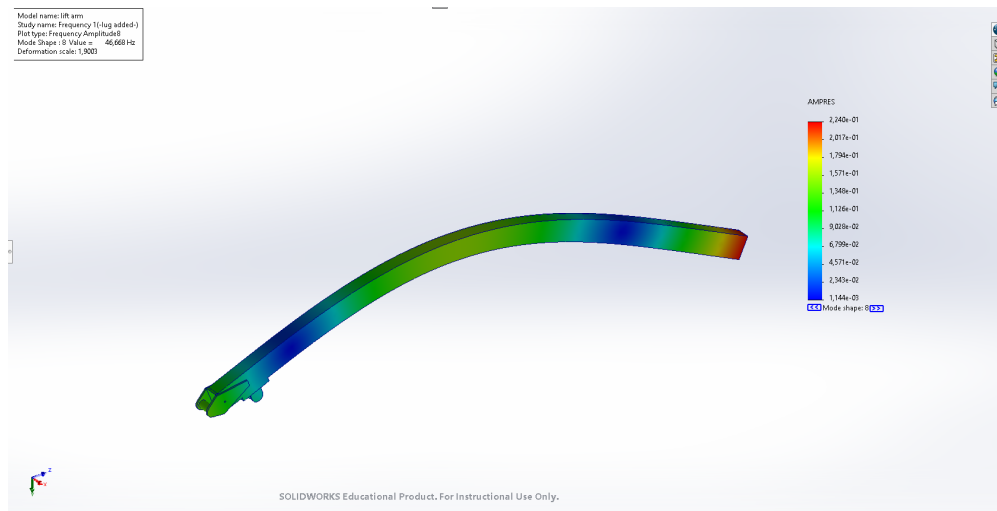


Figure 19: Mode 8 with 46,67 Hz

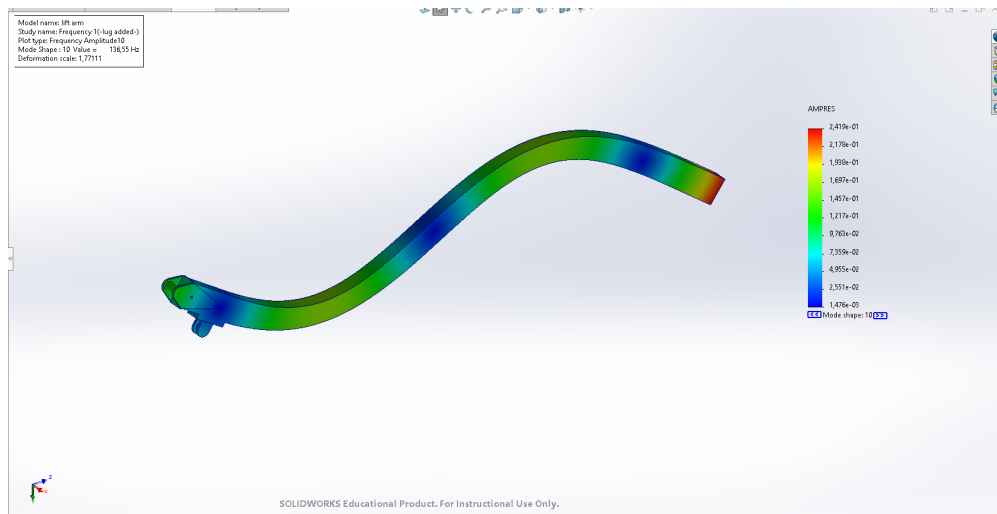


Figure 20: Mode 10 with 136,55

Mode 5 and Mode 8 represent realistic excitation of the system, while Mode 10 corresponds to exaggerated deformation at higher frequency.

3.2 Open Loop: System Response

Open-loop system uses experimental data (time and input data, U_{in} to provide a movement to the mechanism. Figure 22 represents the target trajectory. Figure 21 show how the simulated piston actual movement. The signal is dominated by high frequency oscillation at first and in the end. This could be due to underdamping and it shows that it has excitation as the valve opens and closes.

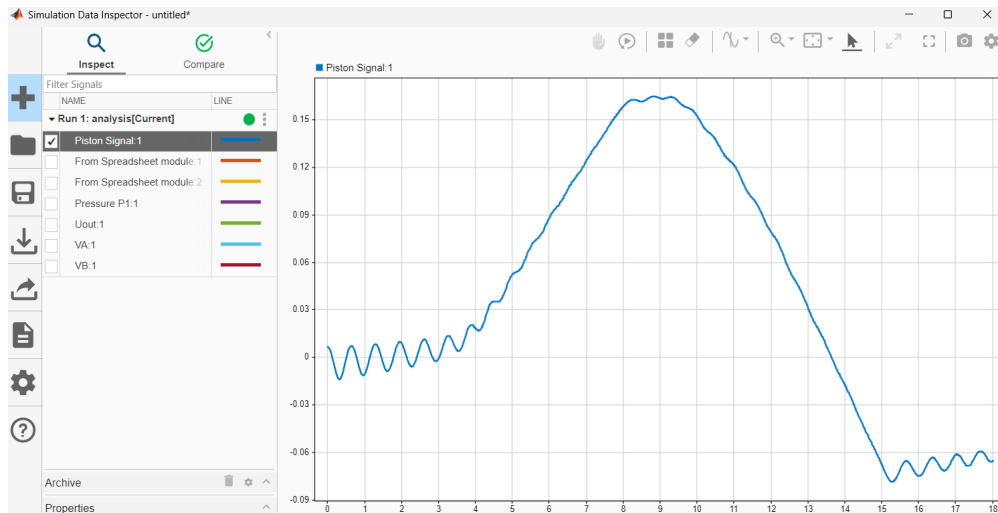


Figure 21: Actual Piston Movement from open-loop system

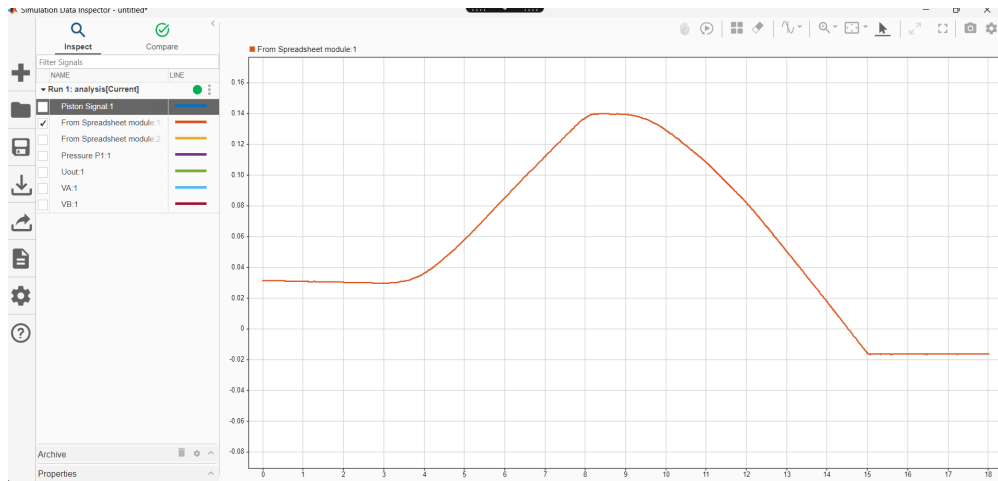


Figure 22: Experimental Piston Data for open-loop system

Figure 23 provides a raw sensor output data. It reveals that mechanically struggling or jerking. It may be due to not having enough force to overcome the static friction.

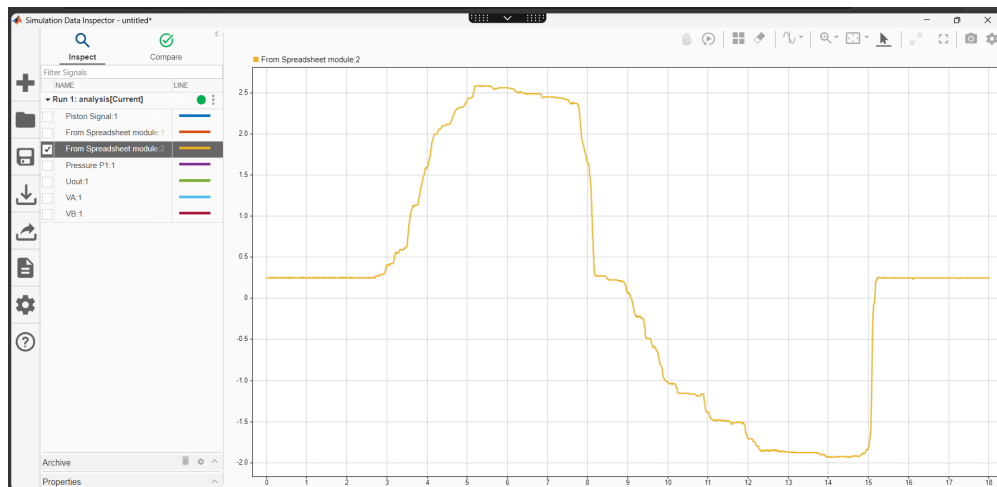


Figure 23: Control Signal Output Voltage from open-loop system

3.3 Closed Loop: System Response

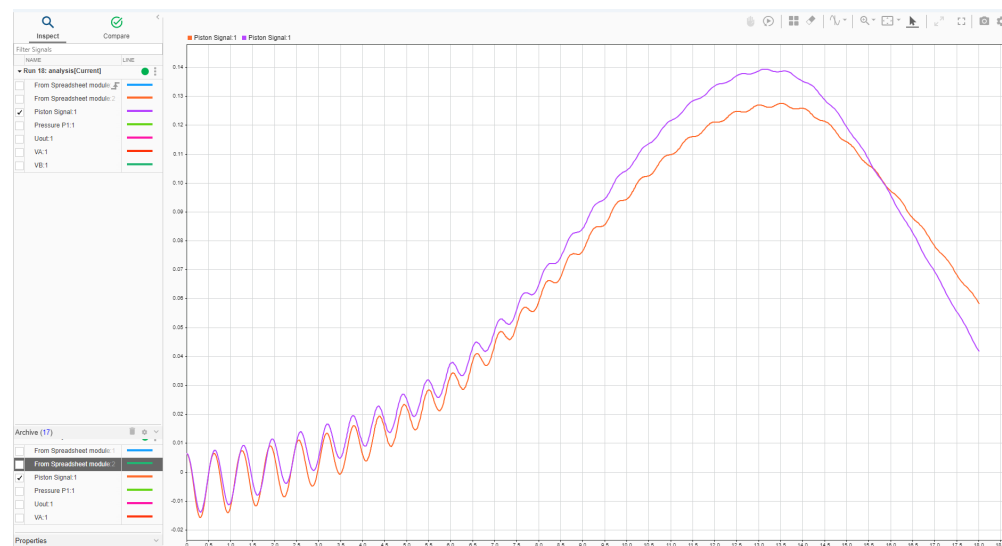


Figure 24: Closed Loop Piston Movement: Purple is tuned and Yellow is Original

The transition from open loop to a controlled system was achieved through implementation of a closed loop PID controller. As shown in Figure 24, the untuned response (yellow) fails to reach to $0.14m$ which is the target from the experimental data. In contrast, the tuned response (purple) demonstrates superior tracking capability and also reaching the amplitude. It also shows that purple has faster rise time meaning the tuned controller is more responsive to the input signal. The tuned version maintains its energy throughout the cycle better than the orange line which looks slightly more damped in comparison.

4 Discussion

The first phase of the hydraulically driven crane model has been successfully completed. By modeling the components in SolidWorks, setting up their properties and coordinates correctly and exporting into

MATLAB, the system effectively maintains 1 DOF. The system also confirms that there is no redundant constraints. This indicates a clean kinematic movement of the system as desired. The model joint coordinates are also verified providing the foundation of the system for future analysis in the next milestone.

4.1 Comparison of Open-Loop and Closed-Loop Performance

The key phase of the overall report is to evaluate the transition from open-loop baseline behavior to a controlled closed-loop state.

- Open-loop Response: In the early phase, the system illustrates an uncontrolled drift from its starting equilibrium of $-0.07m$ to approximately $0.06m$. This was caused by raw U_{in} signal from the experimental data provided by the instruction, acting on the hydraulic without feedback.
- Closed-Loop (Tuned) Response: A tuned PID controller improved the overall performance. By comparing the actual piston data with the experimental data, reaching up to the target of $0.12m$.

By evaluating the provided graphs, the closed-loop (Tuned) system is a significant improvement over both the Open-Loop response. The closed-loop system provides a better controllability

- Oscillation Control: in the open-loop plot, the oscillation is not even in the pattern or movement which is due to the natural frequency of the system. In the closed-loop plot, the PID controller is actively managing the valve to force the piston to follow the reference signal.
- Error Correction: Closed-loop can provide compensation for external disturbances like a change in pressure where open-loop just simply drift.

The transition to closed-loop control the system's behavior more effectively by using "linearizing". Using the **Derivative "D"** component of the controller provides as a digital damper to reduce a high frequency oscillation which can be seen in the open-loop results. The comparison confirms that while open-loop system is useful for identifying the mechanical characteristics such as natural frequency, it is not useful for high precision applications. The closed-loop provides a system with exact target regardless of mechanical non-linearities or external disturbances. The PID parameters provides the simulation to mimic the smooth experimental profile by proving the active feedback which is essential to transform the underdamped mechanism.

5 Conclusion

The first milestone of the design of the control algorithm of the hydraulically driven crane has been achieved by successfully modeling and assembling using SolidWorks and MATLAB SIMULINK. The working mechanism establishes a reliable baseline for the future tasks. The comparison between open-loop and closed-loop performance provides the mechanism is prone to oscillations and by using PID controller, it can effectively compensates for these nonlinearities. The successful tracking of the experimental data of piston confirms that the current model provides a high representation of the physical crane. The working mechanism provides a reliable and validated baseline for the future tasks such as implementing the more advanced control strategies or optimizing the system using various loading conditions.

A APPENDIX: Detailed Drawings

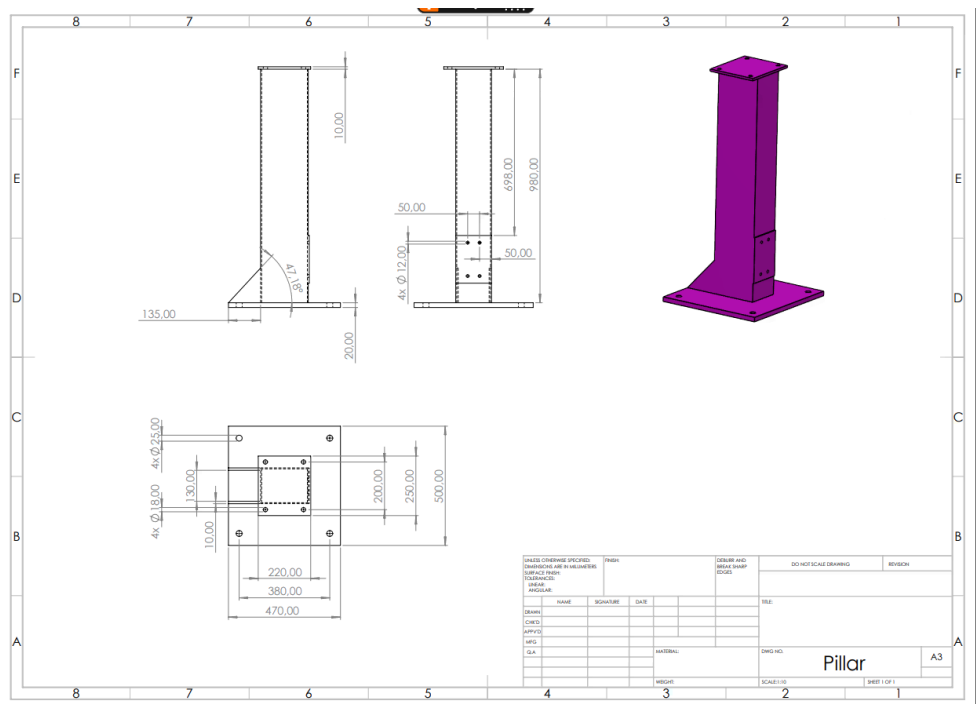


Figure 25: Drawing of Pillar

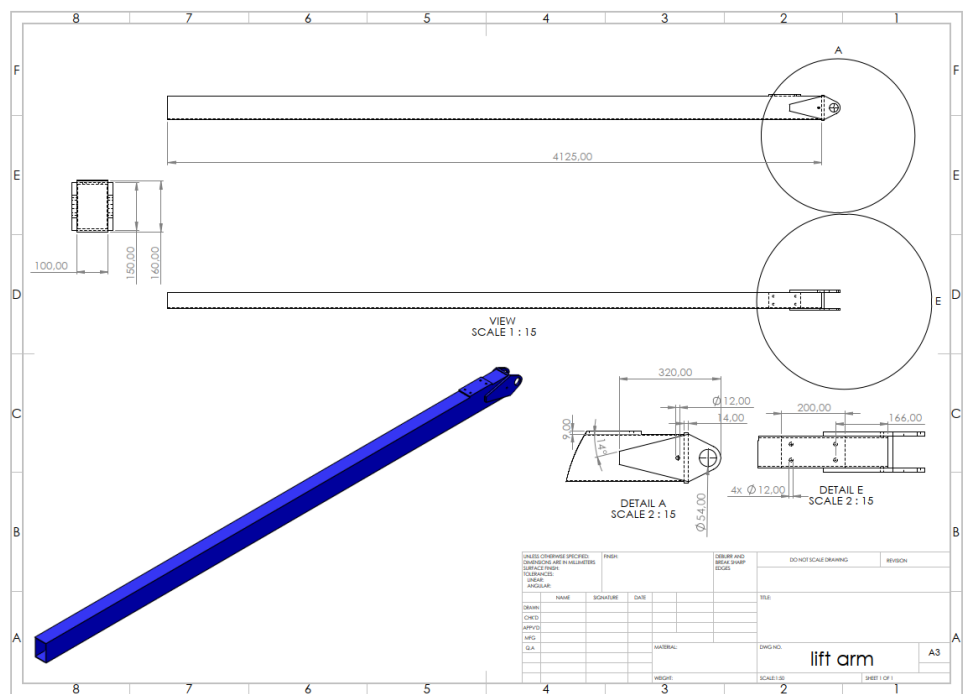


Figure 26: Drawing of the Lift Arm

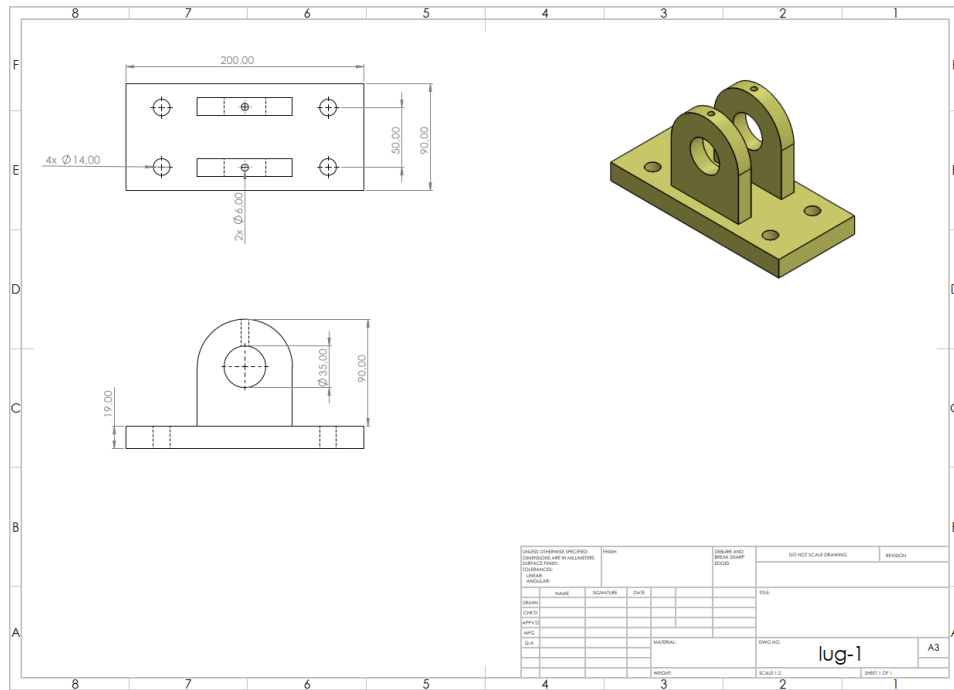


Figure 27: Drawing of Lug 1

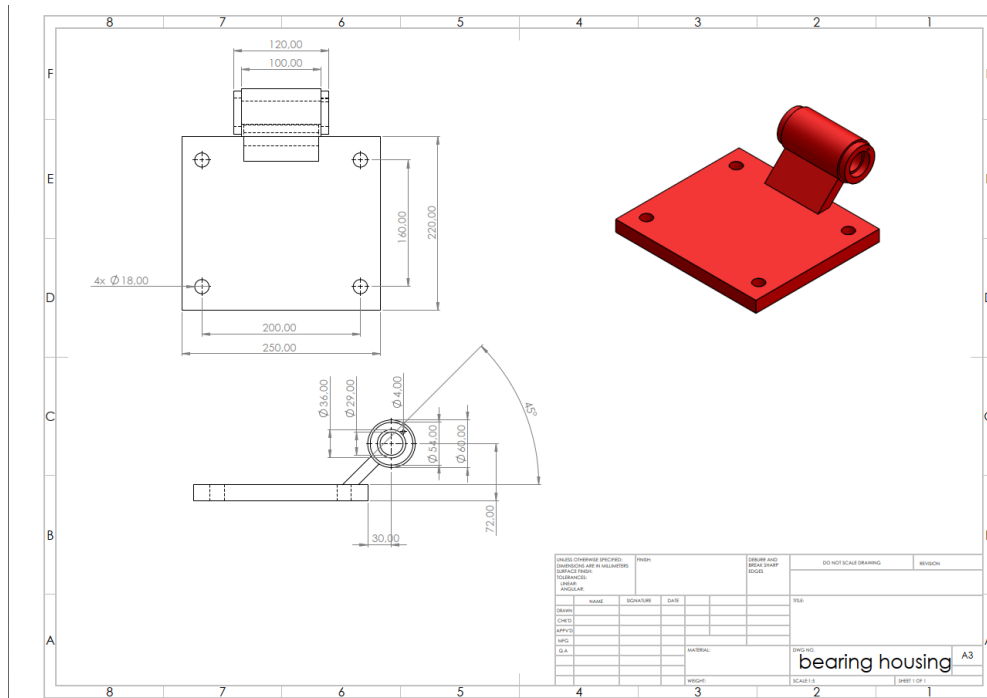


Figure 28: Drawing of Bearing housing

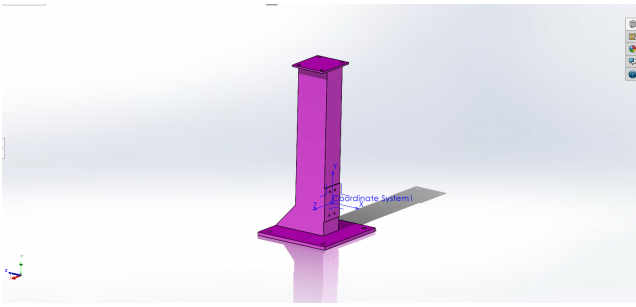


Figure 29: Isometric View of the Pillar with new coordinate

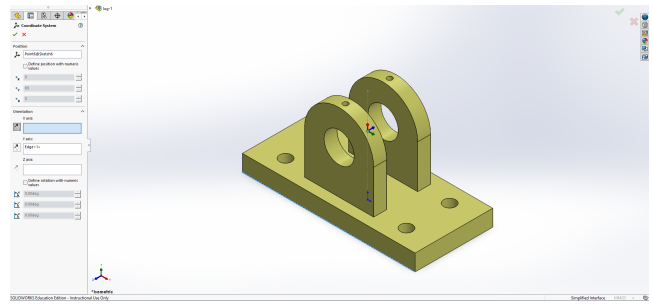


Figure 30: Creating New Coordinate at Preferred Location

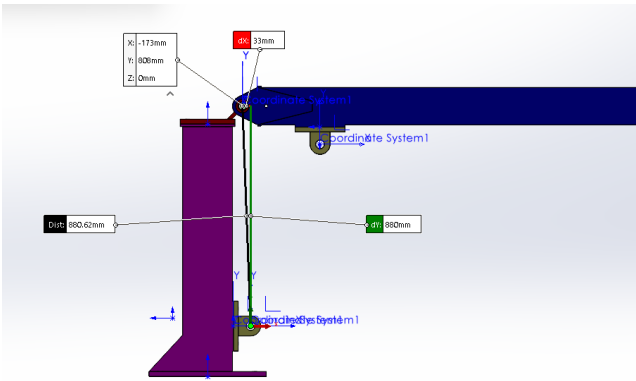


Figure 31: Joint 1 to Joint 2 Coordinate measurement

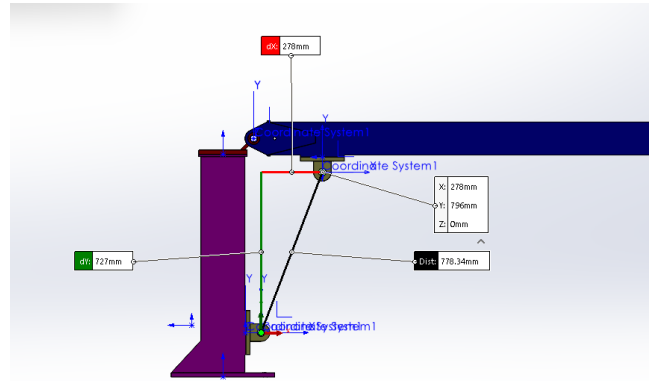


Figure 32: Joint 1 to Joint 3 Coordinate measurement

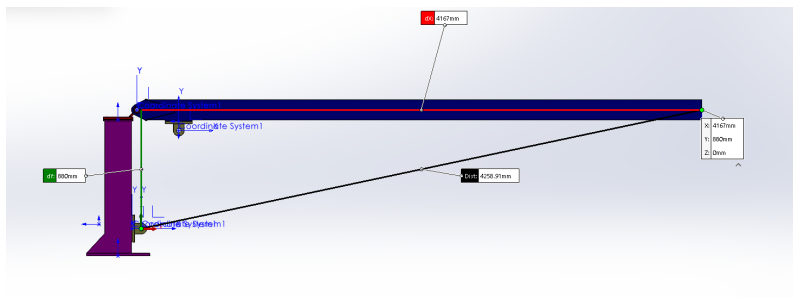


Figure 33: Joint 1 to Joint 4 Coordinate measurement

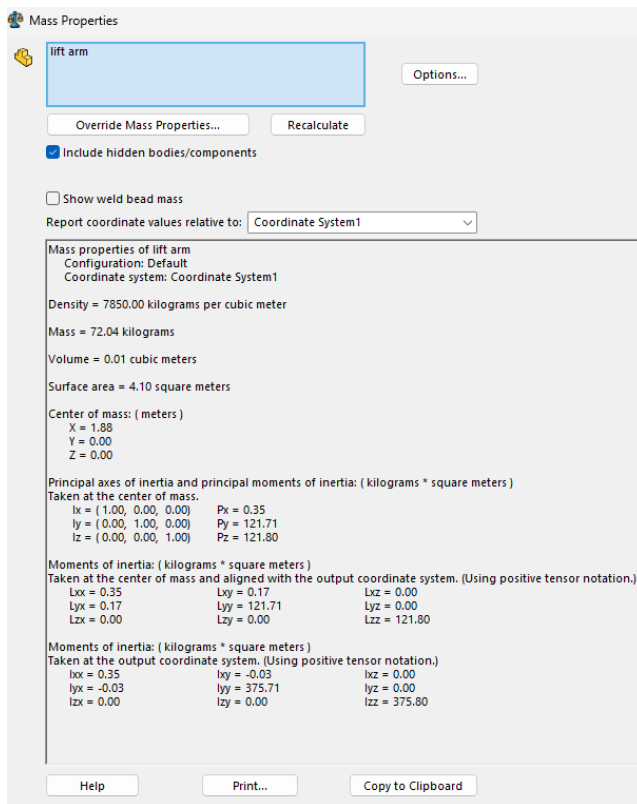


Figure 34: Mass Properties in SolidWorks

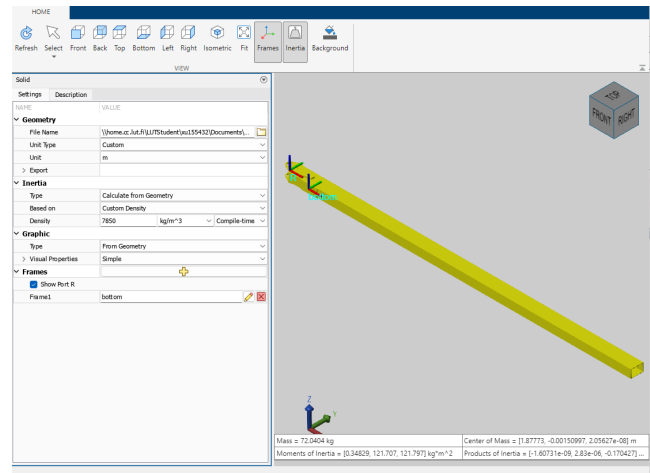


Figure 35: Inertia in MATLAB

B Custom Simscape Component Code

B.1 Custom Directional Valve Component (Simscape)

Listing 1: Custom Simscape Component for Directional Valve

```

1 component custom_component_D_valve
2 % This is .ssc file template for defining a custom simscape component
  block
3 % Nodes are connected to other blocks using the foundation domains.
4 % domainless inputs and outputs are Simscape signal type.
5
6 % Variables include units in them, use '1' if variable has no units,
7 % value(var,'unit') can be used to strip units from variable and
8 % {var, 'unit'} to define units for an unitless variable.
9
10 nodes
11     P = foundation.isothermal_liquid.isothermal_liquid; % P:left
12     T = foundation.isothermal_liquid.isothermal_liquid; % T:left
13     A = foundation.isothermal_liquid.isothermal_liquid; % A:right
14     B = foundation.isothermal_liquid.isothermal_liquid; % B:right
15 end
16
17 inputs
18     Uin = {0, 'V'}; % Uin:left

```

```

19 end
20
21 outputs
22     U = {0, 'V'}; % U:right
23 end
24
25 parameters
26     Cv = {1.0690e-08, 'm^3/(s*V*Pa^0.5)'}; % semi-empirical flow-rate
        constant
27     Tau = {1e-4, 's'}; % Time constant of valve
28     rho = {850, 'kg/m^3'}; % Density of Fluid
29 end
30
31 variables
32     p_PA = {0, 'Pa'}; % pP-pA
33     p_PB = {0, 'Pa'}; % pP-pB
34     p_BT = {0, 'Pa'}; % pB-pT
35     p_AT = {0, 'Pa'}; % pA-pT
36
37     mdot_PA = {0, 'kg/s'}; % mass flow rate from P to A
38     mdot_AT = {0, 'kg/s'}; % mass flow rate from A to T
39     mdot_BT = {0, 'kg/s'}; % mass flow rate from B to T
40     mdot_PB = {0, 'kg/s'}; % mass flow rate from P to B
41 end
42
43 branches
44     mdot_PA: P.mdot -> A.mdot;
45     mdot_BT: B.mdot -> T.mdot;
46     mdot_PB: P.mdot -> B.mdot;
47     mdot_AT: A.mdot -> T.mdot;
48 end
49
50 equations
51     U.der == (Uin-U)/Tau;
52
53     p_PA == P.p-A.p;
54     p_AT == A.p-T.p;
55     p_PB == P.p-B.p;
56     p_BT == B.p-T.p;
57
58     if U>0
59         mdot_PA == rho*(Cv*U*sign(p_PA)*sqrt(abs(p_PA)));
60         mdot_BT == rho*(Cv*U*sign(p_BT)*sqrt(abs(p_BT)));
61         mdot_PB == 0;
62         mdot_AT == 0;
63
64     elseif U<0
65         mdot_PA == 0;
66         mdot_PB == -rho*(Cv*U*sign(p_PB)*sqrt(abs(p_PB)));
67         mdot_AT == -rho*(Cv*U*sign(p_AT)*sqrt(abs(p_AT)));

```

```

68         mdot_BT == 0;
69
70     else
71         mdot_AT == 0;
72         mdot_PA == 0;
73         mdot_PB == 0;
74         mdot_BT == 0;
75     end
76 end
77 end

```

B.2 Custom Double-Acting Cylinder Component (Simscape)

Listing 2: Custom Simscape component for the double-acting cylinder (.ssc)

```

1  component custom_component_template
2  % This is .ssc file template for defining a custom simscape component
   block
3  % Nodes are connected to other blocks using the foundation domains.
4  % domainless inputs and outputs are Simscape signal type.
5
6  nodes
7      A = foundation.isothermal_liquid.isothermal_liquid; % A:left
8      B = foundation.isothermal_liquid.isothermal_liquid; % B:right
9  end
10
11 inputs
12     x    = {0.3, 'm'}; % x:left
13     xdot = {0, 'm/s'}; % xd:left
14 end
15
16 outputs
17     F = {0, 'N'}; % F:left
18 end
19
20 parameters
21     A1 = {pi*(0.025)^2-pi*(0.015)^2, 'm^2'}; % piston-side area
22     A2 = {pi*(0.025)^2, 'm^2'}; % cap-side area
23     V0 = {1.0688e-4, 'm^3'}; % dead volume
24     B_oil = {1.5e9, 'Pa'}; % bulk modulus of oil
25     B_cyl = {2.1e10, 'Pa'}; % bulk modulus of
        cylinder
26     n = {0.9, '1'}; % cylinder efficiency
27     rho = {850, 'kg/m^3'}; % fluid density
28     l = {1.5, 'm'}; % cylinder length
29 end
30
31 variables
32     pA = {63e5, 'Pa'}; % pressure on rod side

```



```

33     pB    = {28e5, 'Pa'};           % pressure on cap side
34     mdotA = {0, 'kg/s'};           % mass flow rate at A
35     mdotB = {0, 'kg/s'};           % mass flow rate at B
36
37     Bh2   = {150, 'Pa'};           % hose bulk modulus (A side)
38     Bh3   = {100, 'Pa'};           % hose bulk modulus (B side)
39
40     Be2   = {1.129e9, 'Pa'};       % effective bulk modulus (A side)
41     Be3   = {1.163e9, 'Pa'};       % effective bulk modulus (B side)
42 end
43
44 branches
45     mdotA: A.mdot -> *;
46     mdotB: B.mdot -> *;
47 end
48
49 equations
50     Bh2 == 10e5 * sqrt(abs(pA)) * {1, '(Pa)^0.5'};
51     Bh3 == 10e5 * sqrt(abs(pB)) * {1, '(Pa)^0.5'};
52
53     let
54         VA = if (A1*x) <= V0, V0 else A1*x end;
55         VB = if (A2*(1-x)) <= V0, V0 else A2*(1-x) end;
56     in
57         Be2 == 1/((1/B_oil)+(V0/((VA)*Bh2)) + ((A1*x)/((VA)*B_cyl)));
58         Be3 == 1/((1/B_oil)+(V0/((VB)*Bh3)) + ((A2*(1-x))/((VB)*B_cyl)));
59
60         mdotA == rho*((pA.der * VA)/Be2) + A1*xdot;
61         mdotB == rho*((pB.der * VB)/Be3) - A2*xdot;
62     end
63
64     pA == A.p;
65     pB == B.p;
66
67     F == ((A1*pA) -(A2*pB))- xdot/{1, 'm/s'}*abs(((A1*pA)-(A2*pB))*(1-n))
68     ;
69 end
end

```

B.3 Reduced-Order Flexible Boom MATLAB

Listing 3: Custom Simscape component for the double-acting cylinder

```

1 stlFile = 'Z:\xu155432\Documents\simulationlaboratory\lift_boom2.STL';
2 figure
3 trisurf(stlread(stlFile))
4 axis equal
5
6 % Material properties

```

```

7 E = 200e9;
8 nu = 0.26;
9 rho = 7800;
10
11 % Interface frame locations
12 origins = [0 0 0;
13             3.6 0 0;
14             0.31 -0.15 0];
15 numFrames = size(origins,1);
16
17 % Create FEM model
18 feModel = createpde('structural','modal-solid');
19 importGeometry(feModel,stlFile);
20
21 structuralProperties(feModel,...
22     'YoungsModulus',E,...
23     'PoissonsRatio',nu,...
24     'MassDensity',rho);
25
26 generateMesh(feModel,...
27     'GeometricOrder','quadratic',...
28     'Hmax',0.1,'Hmin',0.01);
29
30 % Apply boundary conditions
31 faceIDs = [37,39,47];
32
33 for k = 1:numFrames
34     structuralBC(feModel,...
35         'Face',faceIDs(k),...
36         'Constraint','multipoint',...
37         'Reference',origins(k,:));
38 end
39
40 % Reduced order model
41 rom = reduce(feModel,'FrequencyRange',[0 1e3]);
42
43 arm.P = rom.ReferenceLocations';
44 arm.K = rom.K;
45 arm.M = rom.M;
46
47 % Damping
48 dampingRatio = 0.02;
49 arm.C = computeModalDampingMatrix(dampingRatio,rom.K,rom.M);

```